Inequality Analysis and Price Prediction of Cryptocurrency Using ML Models

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Abstract

With the emerging breakthroughs in the ML and AI technologies, the study of cryptocurrency price prediction has emerged as a new area of research. Many articles are research findings produced over time in this field. The behavior of these unrestricted markets with their high volatility, decentralized nature, and liquidity of digital assets make these markets hard to predict using the conventional methods. Even the existing machine learning models will often fail to capture the complexity of these price fluctuations, largely due to the exclusion of many factors such as the features, including market trends and distribution dynamics. This research explores the feasibility of leveraging the machine learning algorithms enhanced by measuring inequality measures such as the **Gini Index** and **Kolkata Index** to improve the accuracy of prediction of cryptocurrencies price.

We begin the project by collecting the price data of the cryptocurrencies (i.e., Bitcoin) from the website called Investing.com, and then we calculate the inequality measures by plotting the **Lorenz curve**. After this, we will measure the statistical measures mean, standard deviation, and kurtosis in order to normalize the data for the training of the models. With the help of these features, we will capture the volatility of price changes, thereby forming a unique feature set for the predictive models. We used various machine learning algorithms, including regression models, decision trees, and random forests, and were trained and evaluated over this new dataset. Finally, we compared the differences between these models under different scenarios.

The results indicate that using the inequality measures has enhanced the predictive performance of machine learning models when compared to the conventional approaches. Finally, this study shows the relationship between wealth distribution in crypto markets and price volatility, helping us to get a great understanding of cryptocurrency price dynamics. The proposed work demonstrates the improvement of short-term price predictions by using the mathematical and statistical measures.

Introduction and Statement of the Problem

The main problem statement and motto of this research study is to focus on analyzing and improving the efficiency in predicting the price of the crypto currencies by incorporating various mathematical methods and using the machine learning models. By leveraging these features such as Gini and K value we will try to address the problem associated with the price volatile nature of the Crypto Currencies. The problems that are mentioned below lead to the study and development of this project.

- What are different machine learning algorithms that can be used to predict the price of the economic markets over the time?
- How accurate are the predictions done by machine learning over the financial markets like crypto currencies?
- What factors do affect the effectiveness of machine learning algorithms when predicting the prices of the crypto currencies?
- How can the existing machine learning models be improved to accurately predict the prices of various Crypto currencies amidst their volatile nature?
- What additional features can be added to train the machine learning models in order to accurately predict the price of the Crypto currencies?
- What are the different inequality measures and how do they describe the Crypto currencies market?
- How can the disparity of price fluctuations and the concentration of wealth in limited subsets be quantified and used to improve cryptocurrency price predictions?

All these questions are answered in our research findings by studying the different mathematical measures such as Gini Index and K value and their effect on the machine learning algorithms when used as features to train them. We studied different cases and compared the accuracies of different models in order to provide a clear insight over different features and dynamics of volatile nature of cryptocurrencies.

Review of Literature

The rise in the popularity of the cryptocurrency market and their high price volatility made cryptocurrencies a popular trading option, and much research has been done over then in order to get insight over their future price to gain profits by trading them. Some of the popular studies over the prediction of the price of cryptocurrencies are as follows,

The research by Xinyi and Wu (2024) shows the prediction of the price of the currencies using different deep learning models, such as long- and short-term model networks. Conventional neural networks. They used different models and approaches, such as Univariate and multivariate, to make it hard to predict the Crypto prices due to their volatility. Their results show that conventional LSTM outperforms remaining models.

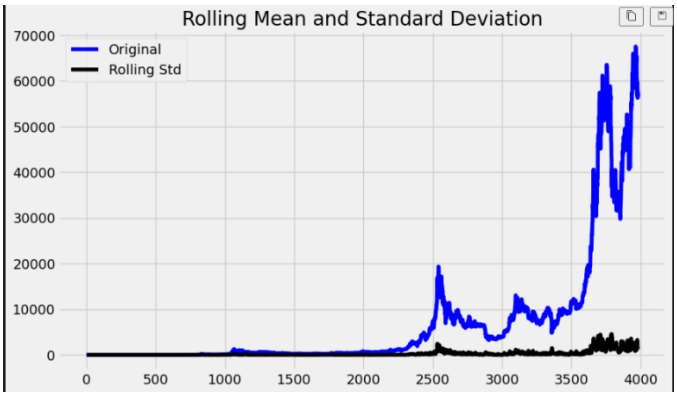
Gudavalli and Kancherla (2023) provided us with valuable information by performing the long-term prediction of the Bitcoin prices using various machine learning techniques. In their research, they highlighted the usage of indicators like RSI, EMA, and SMA, which helps them in studying the price movements and trends with the timeline, thereby enhancing the ability of different machine learning models like LSTM and Random Forest to predict the long-term prices of the Crypto currencies.

Suchismita and Hanesh (2022) have conducted research on the evolutionary dynamics of social inequality in unrestricted competitions. In their study, they mentioned that indices like the Gini Index and K-values have the tendency to converge under unrestricted market systems like the election data, universities, and price markets. They find that these index values tend to stabilize at a point of 0.87 value over the Lorenz curve, which reflects the relationship between the competition and the value of those markets. In our project, we tend to include these mathematical findings over the prices of the Crypto market, as these markets are unrestricted, and we wanted to build the machine learning models to predict the short-term price of cryptocurrencies.

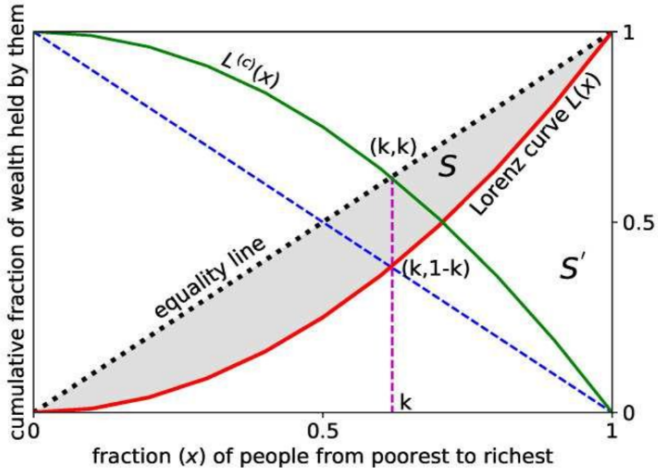
Objectives of the Study

The main objective of this project is to develop a machine learning model that will take advantage of the mathematical analysis of the Crypto currency prices in order to predict the future price of the Crypto currencies. In order to achieve this some the key features we implement in this project are outlined below,

Data Collection: The first primary objective of this project is to gather the comprehensive data that contains the historic price data of the Crypto currency and study the data to identify the treds with it. By examining the data we tried to understand the behavior that might influence the price changes and also the patterns that show the volatility nature of the crypto currencies.



Incorporating Inequality Measures: In this step we will analyze the dynamics of the price in the original data set and calculate the inequality measures by plotting the Lorenze curve over the price feature of the Crypto currency. These measures serve as new features to train the models and will help us to capture the volatility and distribution of prices.



Model Development and Training: In the next step we will make a new dataframe by incorporating the above statistical inequality measures as well the mathematical measures such as standard deviation and kurtosis inorder to train the machine learning models over them. We will develop and train the new machine learning models by splitting this leveraged dataset as train and test splits and will determine the price of crypto currencies using these trained models.

Performance Evaluation: After training different ML models we will evaluate the performance of each of the models over different scenarios such as the lengths of the prediction windows. We will also train the ML models without these leveraged features and will compare their accuracy scores with the leveraged models for a clear understanding of advantages the Inequality measures price predictions.

Data Collection

The dataset utilized in the research work is retrieved from a financial website called investing.com, which provided the historical price data of various cryptocurrencies. From the source, the historic price data of the Bitcoin cryptocurrency is selected, spanning the years between 2012 and 2024, and

is downloaded as a comma-separated (CSV) file. The initial dataset contains features such as date, closing price, opening price, highest price, lowest price, percentage change, and trading volumes.

In addition to these primary features, as discussed above, inequality and statistical measures are introduced to leverage the predictive capabilities of the machine learning models. The leveraged features are as follows, and they will represent their role as mentioned below as,

- **Gini Index**: It represents the inequality within the historical price distribution.
- **K-Value**: It helps us in capturing wealth concentration within price changes over the time.
- **Mean**: It is the average value of the prices.
- **Standard Deviation**: It is the measure to normalize the volatile price distribution.
- **Kurtosis**: It is an indicator that shows the sharp changes within the price distributions.

Unlike other economic entities such as money markets, which are usually restricted and regulated by governmental organizations and agencies, the cryptocurrencies exhibit a unique nature as their prices are unregulated movements driven by market forces. Including the above leveraged features will be crucial in enhancing the ability to predict future prices by inheriting the volatility and price distribution behavior. These derivatives will allow us to capture these characteristics and help us in developing more accurate prediction models to forecast future cryptocurrency prices with higher accuracy.

Exploratory data analysis (EDA) and Hypotheses for the Study

By using the data we imported above, the Lorenz curve is plotted over the calculated closed price difference between the cumulative days. The closed price difference will play a major role here in helping us to determine a window frame to predict the future price. From the dataset, the Lorenz curve here is a plot where the x-axis represents the cumulative proportion of data points, where each data point represents the consecutive days, and the y-axis represents the cumulative proportion of the total value contributed by those data points, which is the closed price differences. If we look over in the code,

The function Lorenz(x) generates the x_axis, which is the cumulative proportion of items for the days, and the y_axis will generate the cumulative proportion of total value, which is the closed difference price. The Lorenz curve starts at (0,0) and ends at (1,1). Over the Lorenz curve, we will plot the perfectly equitable distribution, which is known as the line of equity that is a diagonal line, where y = x. From these lines of equity and the Lorenz curve, the inequality measures will be calculated by leveraging over the areas under the curve.

The Gini Index in the research will help us to understand the inequality between the distribution of prices over time within the cryptocurrency's historical price data.

It is calculated from the Lorenz curve, where if the area between the Lorenz curve and the line of equality is considered as A and the area below the Lorenz curve is considered as B, then the Gini index is calculated as,

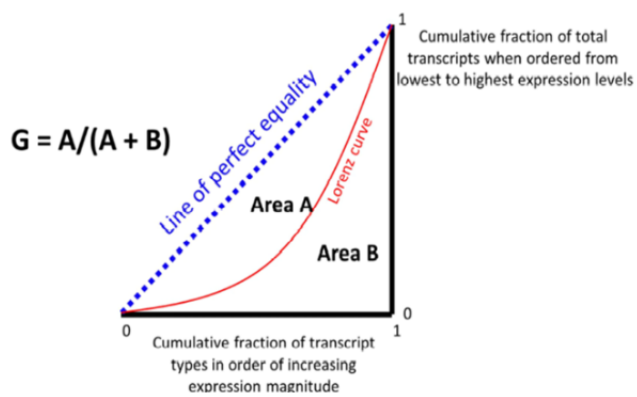
Gini Index = A/(A+B)

To calculate this within the code, we will first find the area under the Lorenz curve by dividing the curve along the axis as the discrete small trapezoids between the consecutive points of the Lorenz curve.

Now, the area is calculated by using the formula below, where we already know the total area under the line of the equity curve along the axis is 0.5 as it forms a triangle of length and breadth 1.

Gini Index = 1-2B

$$B = \sum_{i=1}^{n-1} \frac{(y_{i+1} + y_i)}{2} \cdot (x_{i+1} - x_i)$$

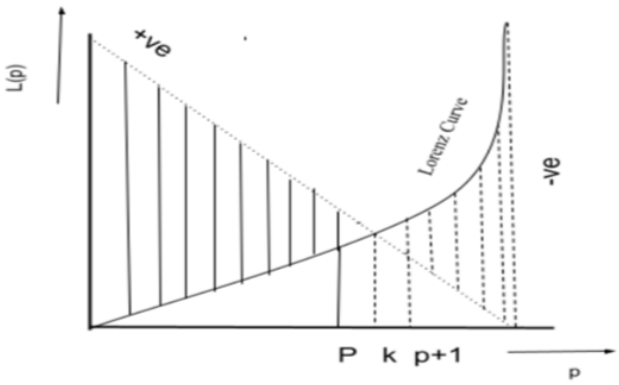


The K-value in the research will help us to understand the concentration of the wealth over the change of price within the period. It is calculated by finding the point where the line perpendicular to the line of equity meets the Lorenz curve.

In the code, this can be calculated for each of the points (x, y) over the Lorenz curve by computing the difference between (1-x) and y. Here the K value is interpreted as the midpoint of the last two X values where the respective difference becomes positive to negative.

-value is ,

$$k = \frac{P_i + (P_i+1)}{2}$$



Data Analytics

From the initial state to the end state of this project we performed many data analytics over the original dataset we downloaded from investing.com in order to compare the performance of the ML model under different scenarios. These data analytics in each step will act as an algorithm for this research in each of the steps.

1. Data Collection and Preparation: In step one which is the initial phase we retrieved bitcoin data for the years 2011–2024 from "**investing.com**". Some of the features within this dataset include date, opening price, closing price, high, low, percentage change, and trading volume.

2. Data Visualization: The next step involves understanding the data inorder generate new ideas to work on it. To do this we created time-series plots to observe historical price trends and volatility of prices. Also plotted the rolling mean and standard deviation to analyze short-term and long-term price movements.

3. Feature Engineering: In the next step based on the understanding from the above visualization we performed analytics over the closed price difference from which we computed the Lorenz Curve, Gini Index, and K-Value to capture price inequality and volatility.

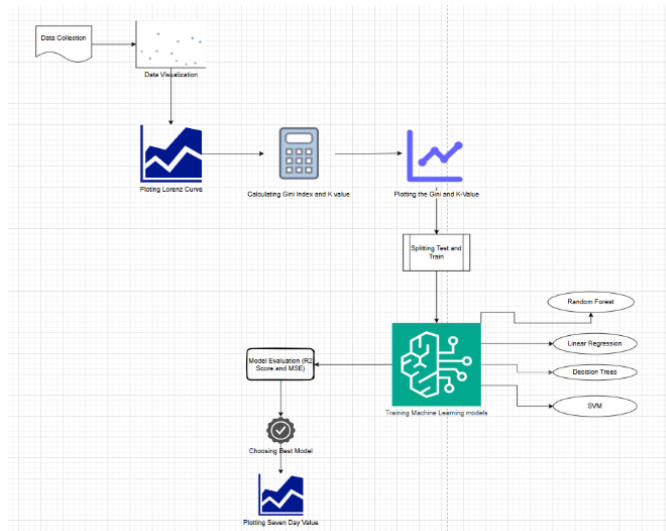
Along with these we calculated the statistical measures such as Skewness, Kurtosis and these steps enhanced the dataset that is used to train the machine learning models and to give a better predictive accuracy.

4. Model Development: This leveraged dataset is used to train the machine learning models including Linear Regression, Random Forest, Decision Trees, and KNN using an (80:20) train-test split.

5. Model Evaluation: After this we performed the Model evaluation using the testing split and the trained models by calculating the R^2 Score and Mean Squared Error (MSE) metrics. The R^2 Score is the measure of variance within the dependent variable that is predicted from the independent variable. Finally, we compared the results of all models to determine the best-performing algorithm.

6. Best Model Selection: Finally we selected Linear Regression as the most effective model, as it is achieving the highest R^2 (0.9993) and lowest MSE (184,970).

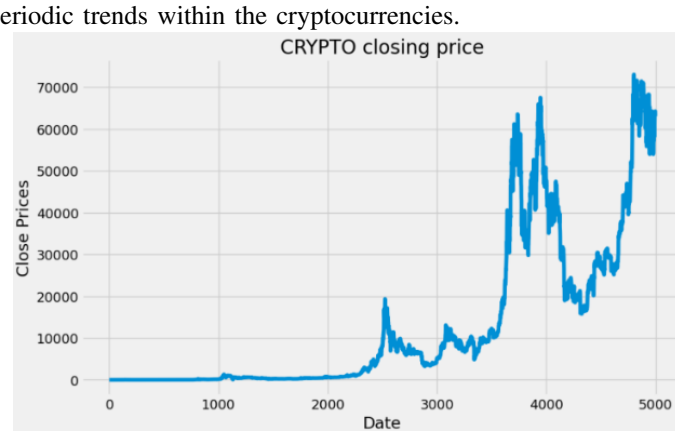
7. Seven-Day Prediction: Predicted prices for the next seven days using the Linear Regression model. Compared results with models trained on basic datasets, demonstrating superior accuracy with engineered features.



Data Visualization and Results Report

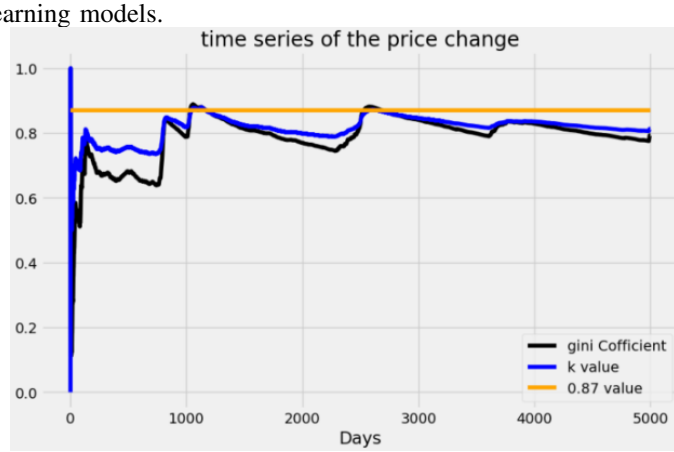
Data visualization plays a vital role in the implementation of this research from the start to the end, it helps us in understanding the complex patterns among the historical prices and their volatile behavior. In this research we will include the visualizations in both the exploratory and evaluating phases, which helps us in deeper understanding of the cryptocurrency data and the performance of the developed machine learning models.

The first plot is the visualization of the time-series plot of bitcoin where in the x-axis we will have the days and in the y-axis we will have the respective prices. This will help us in understanding the historical trends, price fluctuations and periodic trends within the cryptocurrencies.



Next, we plotted the graph of rolling mean and standard deviation, over a rolling moving window inorder to understand how these measures capture the short-term and long-term price movements as shown in the objective of study section.

The Lorenz Curves is plotted using the method mentioned above inorder to visualize and measure the inequality measures within the price distributions. We computed the Gini Index and K-value over the price distribution and plotted them as a line chart along with the 0.87 line to understand the stability of these crypto markets under no restriction and to capture the volatile nature of cryptocurrencies. We utilized these new values as the leveraged features to train our new machine learning models.



In the next step we will split the leveraged dataset into 80% train and 20% test data frames in order to train and evaluate the performance of the machine learning models to predict the upcoming day price.

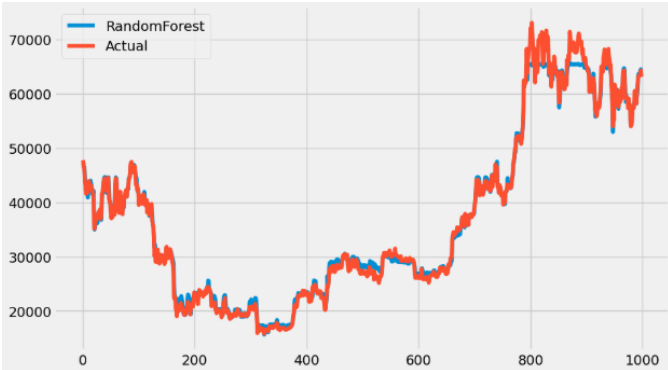
We compared the performance of each of the models over the testing data using the metrics such as Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). All the performances are compared using a line graph for different machine learning models such as the Regression Models, Random Forest, Decision Trees and the KNN model.

The performance of the,

Random Forest Model:

R2 Score Value: 0.9941799102758762

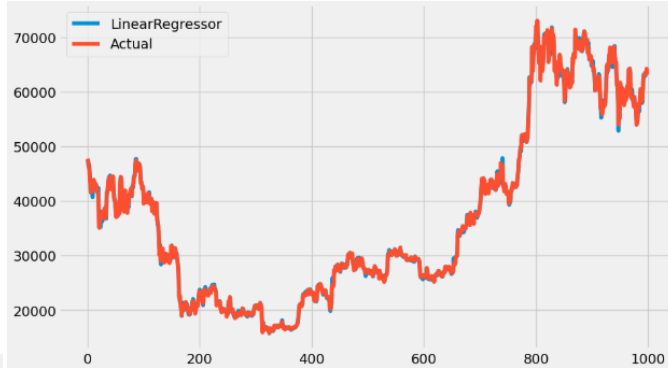
Mean Squared Error: 1547315.9240524005



Linear Regression Model:

R2 Score: 0.999304251004978

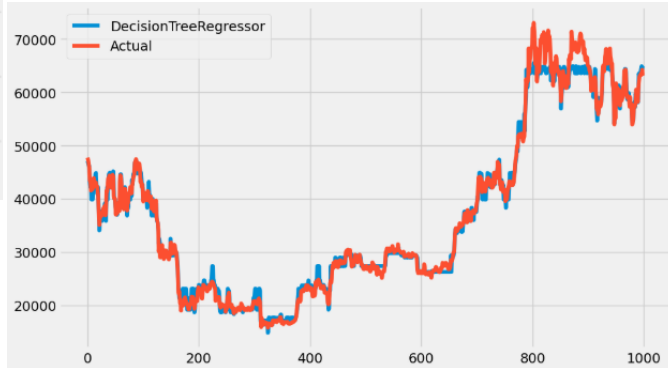
Mean Squared Error: 184970.2582210859



Decision Tree Model:

R2 Score: 0.9905051309027378

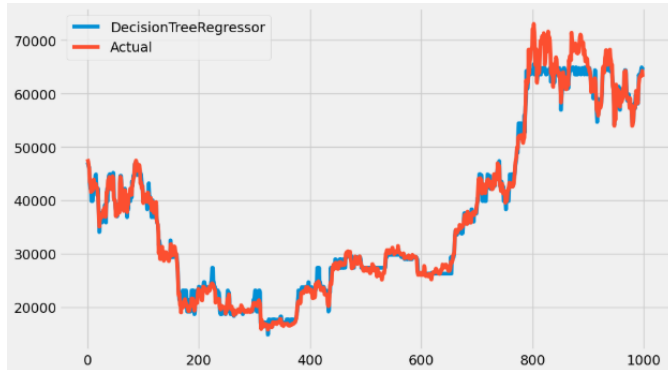
Mean Squared Error: 2524284.4779679687



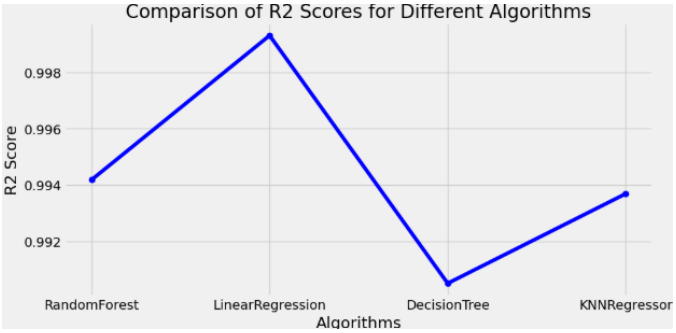
KNN Model:

R2 Score: 0.9936878707487107

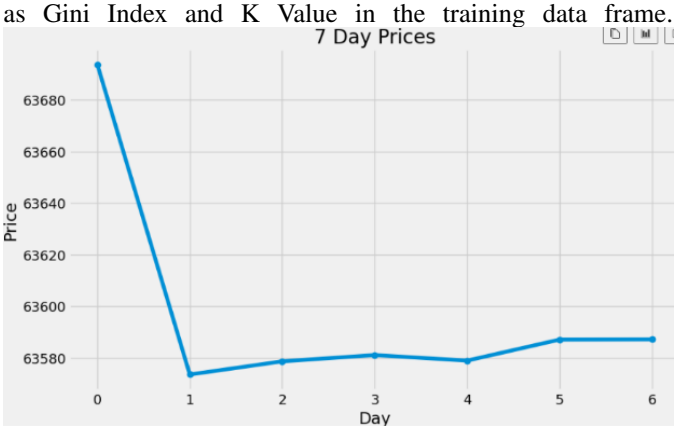
Mean Squared Error: 1678128.4427135154



Then we will have line plots comparing accuracy of all the models. This is useful in selecting the best performing model for seven day prediction. From the above plot we had chosen the Linear Regression model as the best performance model and using it we will find the price for the consecutive seven days respectively.



Finally we will plot the price predictions for the upcoming 7 days as a line bar and compare it with the basic trained models that don't have any leveraged features such as Gini Index and K Value in the training data frame.



From our comparison we found that the models trained over the dataset having the inequality measures and statistical measures had better performance than the models that are trained over the basic original dataset.

Conclusion

In conclusion, by this research we had successfully presented the importance of utilizing the data visualization techniques, inequality measures, and statistical features as leveraged features in training and predicting the prices of unrestricted markets like the cryptocurrency. From the initial part various data visualization helped us in deeper understanding of complex patterns and volatility nature of the crypto prices. The time-series plots, rolling mean and standard deviation graphs, and Lorenz Curves helped in calculation and inclusion of the leveraged features over the historical price data in the machine learning models to enhance their prediction accuracy.

The final phase of the research showed various comparisons of multiple machine learning algorithms, including Regression Models, Random Forest, Decision Trees, and K-Nearest Neighbors (KNN). Performance metrics such as R^2 score and Mean Squared Error (MSE) were calculated to compare the model's efficiency. Among these, the Linear Regression model demonstrated the best performance with an R^2 score of 0.9993 and the lowest MSE of 184,970, making it the optimal choice for future price predictions.

Finally we plotted the predicted prices for the subsequent seven days, and compared these plots with the basic plots to show the correctness and the need for this project. We hope that this project serves as a starting step for further development of models to predict the price of the crypto. Some of the future works can include expanding the dataset to incorporate additional cryptocurrencies. Exploring more advanced machine learning techniques such as ensemble models or deep learning. Incorporating external factors such as macroeconomic indicators, regulatory changes, and social media sentiment analysis. In conclusion, this study provides a detailed roadmap for leveraging data analytics and machine learning to address the challenges posed by cryptocurrency volatility, laying the foundations for more accurate and reliable prediction systems in the unregulated markets.

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